

Case Design for the Mechanism Component

Strategic Symmetry: Arms-Supplier Overlap and the Onset of Militarized Disputes

To: Prof. Paul Poast
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Re: A paired case design: Egypt–Israel and China–Vietnam

1. What the cases are and are not asked to do

The large- N design already carries the identification burden (dyad fixed effects, a non-aligned subsample, a shift-share instrument). The case component is asked to do something different: to establish whether the causal process implied by our theory is *observable* in the decisions of states whose supplier portfolios overlap, and to discriminate among several processes that are all consistent with the same reduced-form correlation.

We organize this as nested analysis in Lieberman’s (2005) sense, which also justifies the case selection. Because the large- N results survive our identification tests, we are licensed to run a model-testing small- N analysis on a case sitting *on* the regression line, and to use a second case sitting *off* it to recover the conditions the specification omits. The framework’s last step is the one we take most seriously: what the off-line case generates goes back into the large- N analysis and is tested there. Section 4 is that return step, not an additional robustness exercise.

On the timing objection to Camp David: we do not ask any case to show that overlap caused peace. We ask whether the supply relationship did observable work in specific crises, which is a question no treaty date can answer by itself.

2. Four candidate mechanisms and what would distinguish them

Mechanism	Logic	Discriminating evidence
M1. Supplier leverage (coercive restraint)	A common supplier can credibly threaten to withhold resupply, spares, or new systems from either party; anticipation raises the expected cost of initiating.	<i>Hoop</i> : at least one documented instance of the supplier conditioning, delaying, or suspending transfers in response to escalatory behavior. <i>Smoking gun</i> : recipient decision-makers invoke the risk of supplier sanction when arguing against escalation.
M2. Sustainment dependence (materiel hostage)	Single-source dependence for spares, munitions, and maintenance caps the war a recipient can actually fight.	<i>Hoop</i> : planning assessments treating resupply as a binding constraint on campaign duration. <i>Caution</i> : M2 predicts restraint on <i>duration and intensity</i> , not onset — if only M2 survives, our DV is mismatched to our mechanism.
M3. Supplier-mediated information	The supplier holds accurate private information about both arsenals and can transmit it, reducing miscalculation about the local balance.	<i>Hoop</i> : supplier demonstrably holds and conveys capability information across the dyad (mediation, force-level annexes, arms-control assurances). This is the mechanism most distinctive to <i>overlap</i> rather than to arms transfers generally.
M4. Side payments (rival)	Overlap is a by-product of a bargain purchased with aid; the pacifying force is money and a political guarantee, not the weapons relationship.	Does leverage operate through the arms channel specifically (delivery timelines, end-use restrictions, congressional notification) or through fungible transfers? Does restraint appear when supply is at stake but aid is not?

M4 is the serious competitor and we treat it as the design's principal rival rather than as a closing caveat. A fifth possibility — professional socialization through common training pipelines — we will code but not treat as a primary hypothesis.

3. The pair: same structure, opposite outcome, one difference

The two cases are chosen so that they share the structural condition the theory cares about — both members of the dyad drawing the bulk of their major weapons from a single dominant supplier — and differ in the outcome. The difference we propose to isolate is not the amount of overlap but the supplier's willingness to condition supply on recipient behavior.

Case A — Egypt–Israel (on-the-line; model-testing). High overlap after 1979 and no new onset. Three features earn it the slot.

1. *Within-case variation in the independent variable.* The Egyptian portfolio moves from Soviet-dominated (overlap near zero) through the 1972 expulsion of Soviet advisers and the 1976 abrogation of the friendship treaty to a US-dominated portfolio through the 1980s.
2. *A competing explanation with a sharp date.* The treaty lets us ask whether restraint at later crisis points tracks the 1979 settlement alone or the deepening supply relationship that followed it.
3. *Within-case observations.* Post-treaty episodes where escalation was plausible and did not occur: the 1981 Sinai withdrawal, the 1982 Lebanon war, the Taba arbitration, the Second Intifada, and the 2011–13 Sinai and Morsi period.

Candidate smoking-gun episodes, to be verified against primary sources: the delay of F-16 deliveries to Israel after the June 1981 Osirak strike and the Beirut raid; the 1982 suspension of cluster-munitions

transfers during the Lebanon war; the sequencing of Egyptian FMS packages against Egyptian conduct through the 1980s.

Case B — China–Vietnam (off-the-line, and inside the region where the null appears). Through the 1990s and 2000s both states drew their major weapons overwhelmingly from Russia — Su-27/Su-30 airframes and Kilo-class submarines to both — so measured overlap is high. Disputes nonetheless recur: the 2011 survey-cable incidents and the 2014 HD-981 rig standoff, which involved ramming and the sinking of a Vietnamese vessel. Russia continued to sell to both throughout, sold Vietnam precisely the systems intended to complicate Chinese operations, and maintained public neutrality on the South China Sea.

This case earns its place for a specific reason: **it is where M1’s hoop test fails.** The supplier possessed leverage — Vietnamese dependence on Russian sourcing was close to total — and there is no evident instance of that leverage being exercised, threatened, or anticipated. If we cannot find one, M1 is not a general mechanism but a conditional one, and the condition is the supplier’s political willingness to attach conditions rather than the recipient’s dependence.

Two things we will not overclaim here. First, normalization in 1991 coincides with the rise in overlap, so the within-case time trend is confounded in the same way Camp David is, and more sharply. The case therefore rests on mechanism evidence at specific crisis points, not on the trend. Second, that trend is if anything *favorable* to our theory — the 1979 war and the 1988 Johnson South Reef clash occur when overlap is near zero, the post-1991 incidents are less severe — and we will say so rather than let it be discovered. What the case establishes is narrower and more useful: at the moments that produced onsets under high overlap, the supplier channel is absent from the observable decision process.

Shadow case: Greece–Turkey. We will keep a short treatment because it supplies the one configuration neither main case does — leverage actually exercised and failing. The 1975–78 US embargo on Turkey is an instance in which the lever was pulled, did not produce restraint, and arguably induced diversification. As a deviation inside a region where our model performs well, it explains an idiosyncratic exception; China–Vietnam explains a systematic regional one, which is what Slide 8 requires.

4. Returning to the large- N : two tests the pair implies

(a) Decompose overlap by supplier. If the pair’s contrast is real, the effect of overlap must depend on who the shared supplier is. Because $\text{Overlap}_{ij,t} = \sum_k \min(w_{ik,t}, w_{jk,t})$ is a sum over suppliers, any partition of the supplier set decomposes it exactly, with no residual, and the constraint that the components share a coefficient is a Wald test against our existing Table 1. We will partition by the supplier’s institutional capacity to condition transfers — proxied by supplier regime type together with statutory export control — rather than by Cold War geography, so that the split operationalizes the theory rather than the bloc structure. The US–Russia contrast in the case pair is exactly this test at $N = 2$.

We flag one hazard in advance. Overlap concentrated in Western suppliers is high precisely for US clients, so a positive result reads as “US clients do not fight each other” — the bloc-proxy objection returning through the back door. The decomposition will therefore be run inside the dyad-FE and non-aligned specifications, not only in the pooled logit, and we will report it as uninformative if it does not survive there.

(b) Weight overlap by import dependence. Building Case B surfaced a measurement problem we think is substantive. Our overlap measure is purely compositional: it records the similarity of two import portfolios and is insensitive to how much of a state’s arsenal is imported at all.

Egypt in 1985 imported nearly everything and imported it from one supplier, so composition and dependence coincide and leverage is real. China after the mid-2000s imports little relative to domestic production, yet the little it imports remains Russian-dominated, so measured overlap stays high while dependence — and any leverage resting on it — does not exist.

The Asia-Pacific is where defense-industrial indigenization is most advanced, which raises the possibility that part of our regional null is a measurement artifact rather than a scope condition. The correction is straightforward: weight overlap by import reliance (arms imports scaled by military expenditure) or interact the two. The prediction is a significant interaction and a narrowing of the regional null. We regard this as the higher-priority item of the two, and we would rather find it ourselves than have a referee find it.

5. Evidence and access

- *Case A*: SIPRI Trade Register; DSCA *Historical Facts Book*; the CRS series on conventional arms transfers to the Middle East; Arms Export Control Act end-use provisions and congressional notification records. For decision-making: *FRUS* Carter and Reagan volumes; the Carter and Reagan Presidential Libraries; National Security Archive Egypt–Israel collections; Israeli State Archives releases.
- *Case B*: SIPRI and Rosoboronexport contract reporting; Russian official statements on the South China Sea; Vietnamese and Chinese procurement analyses; the secondary literature on Russian arms exports and Russia’s position in the South China Sea. Russian and Chinese decision-making is far less accessible than the Camp David record, so Case B will rest on the timing of transfers against the timing of incidents, and on the absence of supply considerations from the observable record, rather than on internal deliberation.
- *Method*: Bennett and Checkel (2015); Beach and Pedersen (2019); Fairfield and Charman (2022); Lieberman (2005) for the nesting; Rohlfing (2008) on its pitfalls.

6. Sequence

Week 1	Import-dependence weighting: build the measure, re-estimate, check whether the regional null narrows. This runs first because it can change what the cases have to explain.
Weeks 1–2	Overlap series for both dyads, 1950–2014, with MID onsets overlaid; verify that China–Vietnam is in fact high-overlap-with-onsets in our own data before any case work begins.
Weeks 2–3	Supplier decomposition and the corresponding instrument.
Weeks 3–5	Episode-level evidence for both cases, coded against M1–M4.
Week 6	Draft the case section (approx. 4,500 words) as Section 6.
Deferred	IV probit / control function; pre-trend checks; the dominant-supplier exercise, now reframed as a placebo rather than a robustness check, since dropping the US removes the mechanism rather than stress-testing it.

7. Three questions for you

1. Does China–Vietnam strike you as strong enough to carry the off-line slot, given how much thinner the decision-making record is than for Camp David? Our judgment is that the absence of supplier considerations is itself the finding, but we may be leaning too hard on an absence.
2. If the import-dependence weighting substantially narrows the Asia-Pacific null, does Case B lose its point — or does it become the case that explains the residual?
3. Should this material be a section of the current paper, or does it carry enough weight to be a companion piece?